

Test HallA Moller Quads operation at 300 Amps

January, 2016

HallA Moller Quads MQO1H02 and MQO1H03A were tested to 300 Amps operating current. Below are the temperature measurements for the magnet coils and power supplies.

Magnet coil temperature measurements:

Temperature sensors provided by the “Magnet Measurement Group” were affixed to the four coils of each MQO magnets; recorded values are shown below.

<u>Current</u>	<u>Temp Coil1 (C)</u>	<u>Temp Coil2 (C)</u>	<u>Temp Coil3 (C)</u>	<u>Temp Coil4 (C)</u>
0	28.9	28.2	30.5	28.4
300 Amps, 1 hour	38.2	36.1	40.6	37.5
300 Amps, 2+ hours	38.0	36.2	40.5	37.5

Table 1. Temperature measurement coil MQO1H02, error + 3/-1 Celsius

<u>Current</u>	<u>Temp Coil1 (C)</u>	<u>Temp Coil2 (C)</u>	<u>Temp Coil3 (C)</u>	<u>Temp Coil4 (C)</u>
0	28.9	28.5	29.9	28.5
300 Amps, 1 hour	42.3	40.5	44.0	42.1
300 Amps, 2+ hours	41.8	40.1	43.9	42.1

Table 2. Temperature measurement coil MQO1H03A, error + 4/-1 Celsius

The coils remained below 50 Celsius at 300 Amps. The temperature interlock to the power supplies was verified to be operational. The temperature switches were not verified.

Power supply cabinet temperatures:

The internal air temperature of the cabinet housing the Moller Quad “Sorenson” supplies was measured as 35 degrees C when operating all three supplies simultaneously at 300 Amps.

Thermal camera images of the internal cabling and lugs are shown below.

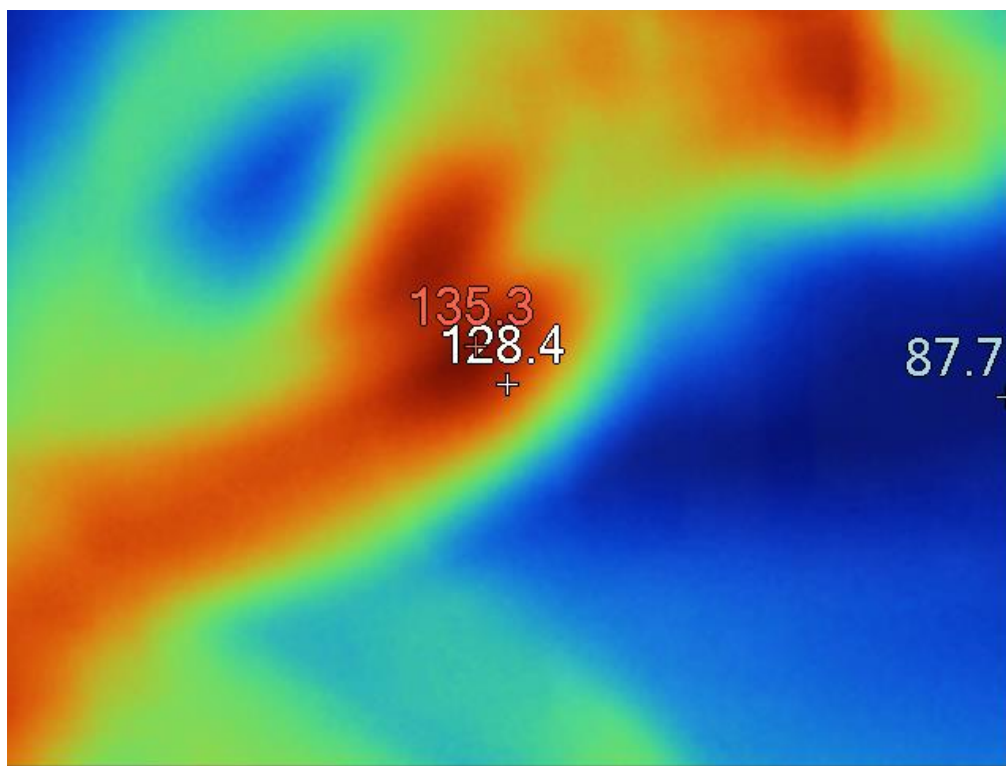


Figure 1. MQO1H02 polarity switch cable temperature (Fahrenheit)

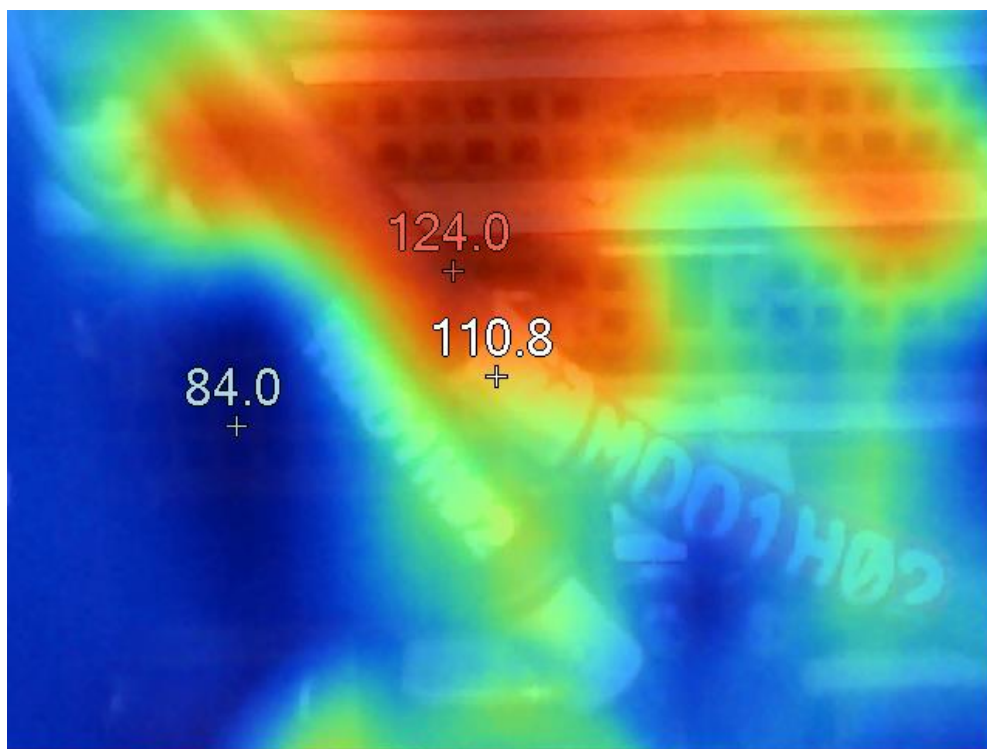


Figure 2. MQO1H02 power cable temperatures (Fahrenheit)

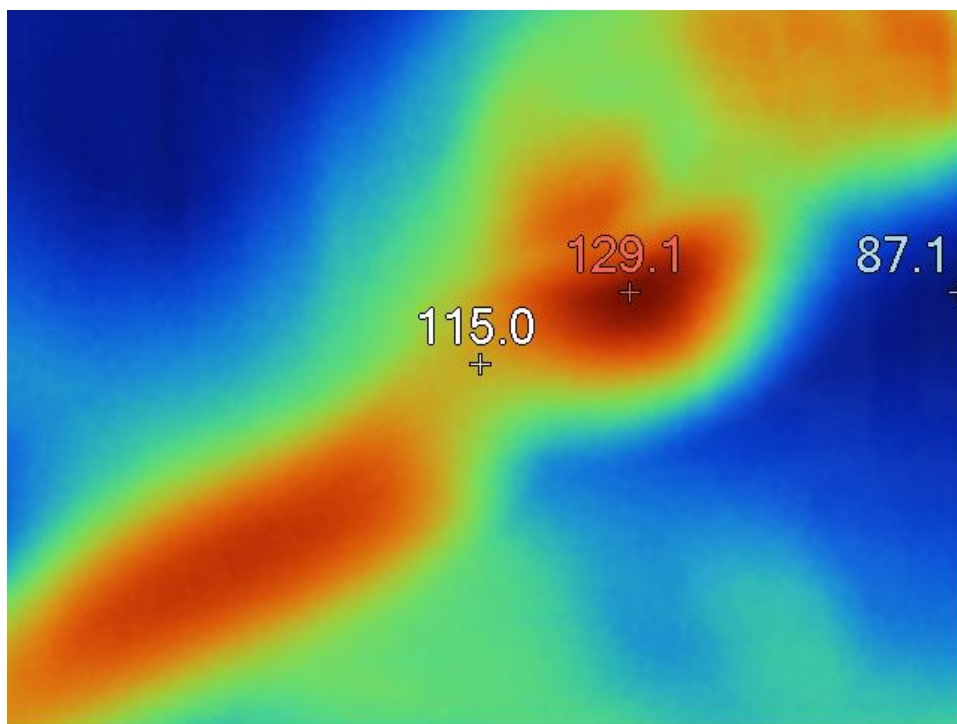


Figure 3. MQO1H03A polarity switch cable temperatures (Fahrenheit)

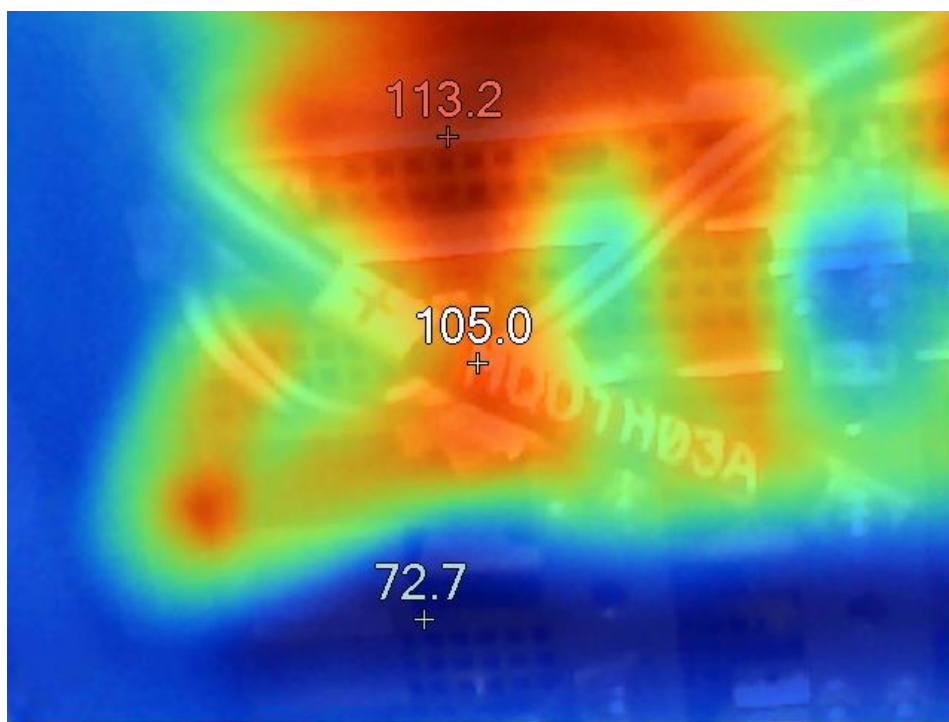


Figure 4. MQO1H03A power cable temperatures (Fahrenheit)



Figure 5. MQM1H02 polarity switch cable temperatures (Fahrenheit)

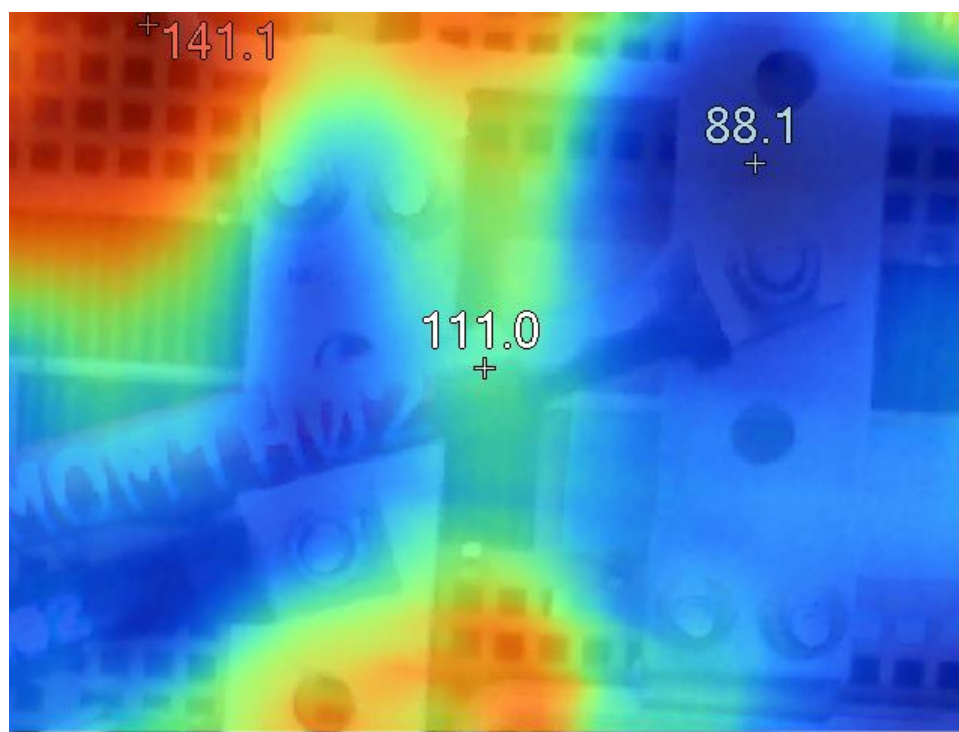


Figure 6. MQM1H02 power cable temperatures (Fahrenheit)

Conclusion:

The maximum measured temperature on the MQO____ coils was around 45 Celsius. The maximum temperature reached on the power cables at 300 Amps is around 55 Celsius.

Based on these measurements it appears operating the MQO quads to 300 Amps should be acceptable.

If permissible, adding a circular cooling fan to the cabinet of the power supplies will allow for better cooling of the power cables.